LITERATURE SURVEY

Prompt Versioning and Regression Testing using LLM-as-a-Judge

1. INTRODUCTION TO LITERATURE SURVEY

Large Language Models (LLMs) are widely used in modern applications to generate text, answer questions, and perform different tasks. However, their performance can change when prompts are modified or when the underlying AI models are updated. Therefore, testing and evaluating AI-generated responses is important.

Several researchers have studied prompt performance, regression testing, and automated evaluation using Large Language Models. The selected research papers explain the problems that occur when prompts or AI models change and how LLM-as-a-Judge can be used to evaluate generated responses.

This literature survey studies two research papers related to our project. The first paper focuses on regression testing for changing LLM APIs, while the second paper focuses on the reliability of LLM-as-a-Judge evaluation. These studies help us understand existing methods, their limitations, and the need for our proposed system.

2. REVIEW OF RESEARCH PAPER 1

Paper Title: (Why) Is My Prompt Getting Worse? Rethinking Regression Testing for Evolving LLM APIs

Authors: Wanqin Ma, Chenyang Yang, and Christian Kästner.

Publication: IEEE/ACM 3rd International Conference on AI Engineering (CAIN), 2024.

Literature Review:

This research paper explains how the performance of AI applications can change when Large Language Model APIs are updated. Sometimes, developers update their prompts to improve the output, but these changes may reduce the accuracy of previously working prompts.

The researchers conducted experiments using different GPT-3.5 models and different prompt designs. They tested these prompts using two datasets related to toxicity detection. They compared the results using Accuracy and F1-score.

The study found that some prompts performed better on older models but gave worse results on newer models. It also showed that different prompt designs can produce different results even when used for the same task.

The researchers suggested that developers should maintain prompt versions and regularly perform regression testing whenever prompts or AI models are updated.

Limitations:

- The experiments mainly focused on toxicity detection.
- The study used specific GPT-3.5 models and datasets.
- AI-generated outputs can sometimes be inconsistent.
- More research is required for testing different types of generative AI applications.

Relevance to Our Project:

This paper is directly related to our project because it explains the importance of maintaining different prompt versions and checking performance changes. We use this idea to develop a system that compares prompt versions and identifies performance improvements or reductions.

Paper-1.pdf

3. REVIEW OF RESEARCH PAPER 2

Paper Title: Judging LLM-as-a-Judge: Concerning Rubric Artifacts in LLM-based Automated Text Generation Evaluation

Authors: Anshul Bagaria, Sowmya S. Sundaram, Gokul S. Krishnan, and Balaraman Ravindran.

Publication: arXiv research paper, 2026.

Literature Review:

This research paper focuses on using one Large Language Model to evaluate the responses generated by another AI model. This method is called LLM-as-a-Judge. It is used to check the quality of AI-generated answers based on evaluation criteria called rubrics.

The researchers studied whether an AI judge gives fair and reliable scores when evaluating generated responses. They conducted experiments using HEALTHBENCH and RESEARCHRUBRICS datasets. They used different AI models and tested whether the evaluation results were affected by the wording of the rubrics.

The study found that an AI judge can sometimes produce unreliable results. It may depend too much on the evaluation instructions instead of properly checking the actual answer. In one experiment, when the generated response was changed to reverse its meaning, the judge gave the expected changed decision in only 37.7% of valid test pairs.

Limitations:

- The experiments mainly focused on rubric-based evaluation.
- The study used specific datasets and AI models.
- The results may not apply to every domain or language.
- More research is required to improve the reliability of AI-based evaluation systems.

Relevance to Our Project:

This paper is useful for our project because we use LLM-as-a-Judge to evaluate responses generated by different prompt versions. It helps us understand that AI judges may also make mistakes. Therefore, our system should use clear evaluation criteria and compare evaluation results carefully to improve reliability.

Paper-2.pdf

4. REVIEW OF SUPPORTING RESEARCH

Apart from the two main research papers, some additional research works are useful for understanding our project.

1. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena (2023)

- Authors: Lianmin Zheng and others.
- This research explains how one AI model can evaluate responses generated by other AI models.
- It studies the use of AI judges for comparing response quality.
- It provides a foundation for using LLM-as-a-Judge in our project.

2. Prompt Exploration, Perturbation, Testing and Iteration using Visual Analytics for Large Language Models (PromptAid, 2023)

- Authors: Aditi Mishra and others.
- This research focuses on exploring and modifying prompts.
- It helps users understand how prompt changes affect AI-generated outputs.
- It is useful for understanding prompt improvement and testing.

3. Pre-Train, Prompt, and Predict: A Systematic Survey of Prompting Methods in Natural Language Processing (2023)

- Authors: Pengfei Liu and others.
- This research explains different prompting methods used in language models.
- It discusses how prompt design affects the performance of AI models.
- It provides background knowledge for our prompt versioning system.

These supporting works are cited in the reference list of the first and second uploaded research papers.

Paper-1.pdf

Paper-2.pdf

5. COMPARISON OF EXISTING APPROACHES

The two main research papers focus on different parts of our project.

Comparison	Research Paper 1	Research Paper 2
Main Topic	Regression testing	LLM-as-a-Judge
Main Focus	Prompt and model changes	AI evaluation reliability
Method	Different prompts and GPT models	Rubric-based evaluation
Testing	Accuracy and F1-score	Classification and counterfactual tests
Main Problem	Performance reduction after updates	Unreliable evaluation decisions
Project Connection	Prompt version comparison	AI-based response evaluation

Overall Comparison:

The first paper explains why prompt versioning and regression testing are necessary.
The second paper explains why AI-based evaluation must be checked for reliability.
Both papers provide important ideas for developing our project.

Paper-1.pdf

Paper-2.pdf

6. RESEARCH GAP

From the study of the selected research papers, we identified the following gaps:

- Existing research explains the importance of prompt versioning, but there is a need for a simple system to store and manage multiple prompt versions.
- Changes in prompts can affect AI-generated responses, but developers need an easier way to automatically compare old and new prompt performance.
- LLM-as-a-Judge can evaluate AI-generated responses, but its evaluation results may sometimes be unreliable.
- There is a need to combine prompt version management, regression testing, and AI-based evaluation in a single system.
- Developers need a simple interface to view evaluation scores, compare prompt versions, and identify the best-performing prompt.

Our Proposed Solution:

Our project aims to address these gaps by developing a system that stores multiple prompt versions, performs automated regression testing using predefined inputs, and evaluates generated responses using LLM-as-a-Judge. It will compare the results and help users identify which prompt version provides better performance.

7. CONCLUSION

The literature survey helped us understand the importance of prompt versioning, regression testing, and automated evaluation in Large Language Model applications. The first research paper explains how prompt performance can change when AI models are updated. The second research paper highlights the reliability problems that can occur when one AI model evaluates another.

By studying these research works, we identified the need for a system that combines prompt management, automated testing, and AI-based evaluation. Our proposed project aims to make prompt testing easier, compare different prompt versions, identify performance changes, and improve the reliability of AI-generated responses.